# Kalm Language Specification

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|---|---|
| VERSION | 1.1 |
| STATUS | FINAL |
| DATE | 2026-10-10 |
| JAVA VERSION | 25 |

# Contents


- 1. The shape of Kalm
  - A small taste
- 2. Source files and lexical rules
  - Tokens, whitespace, and comments
- 3. Types, values, and Java mapping
- 4. Bindings, expressions, and functions
- 5. Flow, lambdas, and errors
  - Matching and checks
  - Resource and scope management
  - Tasks and synchronization
- 6. Classes, contracts, and visibility
- 7. Packages and Java interoperability
- 8. Standard library
- 9. Core grammar
- 10. Diagnostics
- References


## 1. The shape of Kalm

Kalm is a statically typed, object-oriented JVM language. It has Java's class model, predictable execution, explicit types at API boundaries, and direct access to Java libraries. Everyday syntax uses Java-style braces, Kalm keywords, inferred local types, and concise expression-bodied functions. Blocks use either braces or a colon followed by an inline or indented body. Simple statements end with a semicolon or a logical newline; semicolons are recommended.

Kalm source files use UTF-8 and the `.kalm` extension. A JDK 25 or newer is required to compile Kalm programs; generated programs run on Java 25 or newer.

Kalm does not bundle Java libraries. Any library available to the host Java project is available to Kalm, subject to normal Java visibility, classpath, module readability, and module export rules.

### A small taste

KALM

```
space studio.greetings;
use java.time.LocalDate;
use java.util.ArrayList;

pub class Greeter {
    fun greet(name: Text) -> Text = @"Hey, $name. Glad you're here.";
}

pub fun main(args: Array<Text>) -> Unit {
    fix today = LocalDate.now();
    fix names = ArrayList<Text>();
```

```kalm
    names.add("Tim");
    sayLine(Greeter().greet(names.get(0)));
}
```

Kalm uses Java classes and interfaces, supports direct calls to Java APIs, and provides both brace-delimited and indentation-delimited blocks.

# 2. Source files and lexical rules

A file contains an optional `space` declaration, zero or more `use` declarations, then declarations. Top-level executable statements are not allowed. A source file may contain at most one `pub fun main(args: Array<Text>) -> Unit` or zero-argument `pub fun main() -> Unit` entry point. The compiler must reject multiple selected entry points for one application. The compiler generates the JVM-required `public static void main(String[] args)` launcher; for the argument-taking form, the Kalm entry point receives the original Java `String[]` as `Array<Text>` without copying. A compiler gives each source file with top-level definitions a deterministic Java facade named `<SourceBase>Kalm`. Public top-level classes and contracts live in same-named `.kalm` files.

If `space` is omitted, the file uses the unnamed package. A filename that cannot produce a unique Java identifier for its Java facade is a compile error.

## Tokens, whitespace, and comments

Identifiers are case-sensitive, start with a Unicode letter or underscore, and continue with Unicode letters, digits, or underscores. The lexer recognizes multi-character operators before their prefixes using longest-match tokenization. `@` introduces an annotation use.

Simple statements terminate with `;`, a logical newline, or immediately before the end of a block or file. Separate simple statements on the same physical line require `;`. Semicolons are recommended but optional. A completed `return` followed by a logical newline is a bare return. Expressions continue inside parentheses, indexing, collection literals, and generic type arguments, and after an unfinished operator, comma, or member-access operator.

Every block body may use `{ ... }` or `: ...`. A colon followed by a newline requires a nonempty indented body, terminated by dedentation. Blank and comment-only lines do not affect indentation. Tabs expand to eight-column stops; ambiguous tab/space indentation and inconsistent dedentation are errors. An inline colon suite contains simple statements or declarations separated by semicolons and ends at its logical newline. Nested compound bodies require a multiline suite. Empty bodies use `{}`. Brace blocks and colon blocks may nest freely; indentation is structural only in colon suites. Literal delimiters retain their existing grammar.

Control-flow headers accept either parenthesized or unparenthesized conditions and loop bindings. Colon-bodied lambdas use parenthesized parameters to distinguish the body colon from parameter type annotations. For example:

```kalm
pub fun main() -> Unit {
    fix ready = true
    if ready:
        sayLine("Ready");
    else {
```

```kalm
        sayLine("Waiting")
    }
}
```

The keyword `fixed` is not valid in Kalm 1.1; use `fix`.

Comments use # to end a line and ## `...` ## for a block comment. Documentation comments begin with ### and continue to the end of that line. The lexer recognizes ### first, then the two-character block opener ##, then the single-character line comment #. Block comments do not nest. Documentation comments are preserved as generated Java documentation where practical.

Reserved words are: `space`, `use`, `class`, `contract`, `fun`, `var`, `fix`, `pub`, `internal`, `private`, `protected`, `impl`, `inherits`, `base`, `needs`, `closed`, `extendable`, `singleton`, `data`, `choice`, `alias`, `given`, `generate`, `augment`, `read`, `write`, `lazy`, `late`, `constant`, `forward`, `choose`, `when`, `if`, `elif`, `else`, `while`, `for`, `in`, `next`, `stop`, `return`, `self`, `with`, `defer`, `async`, `await`, `task`, `guard`, `try`, `handle`, `finally`, `raise`, `expect`, `require`, `ensure`, `external`, `bridge`, `annotation`, `target`, `legacy`, `test`, `before`, `after`, `init`, `override`, `true`, `false`, `null`, `and`, `or`, `not`, `as`, `is`, and the built-in type names in Section 3. Java keywords are not Kalm keywords unless listed here, so Java identifiers remain usable in interop.

Text literals use double quotes. Supported escapes are `\\`, `\"`, `\n`, `\r`, `\t`, and `\u{HEX}` for a Unicode scalar value. Interpolated text uses `@"..."`; `$name` inserts a name and `${expression}` inserts an expression. A literal dollar sign is `$$`.

Decimal integer literals have type `Int`, or `Long` with an `L` suffix, and must fit their signed 32-bit or 64-bit range. Floating literals default to `Double`; an `f`/`F` suffix gives `Float`. Numeric separators and hexadecimal or octal integer literals are not part of Kalm 1.1. A character literal uses single quotes and contains exactly one UTF-16 code unit.

```kalm
fix note = @"Today is ${LocalDate.now()}";
fix quoted = "She said: \"hello\"";
```

# 3. Types, values, and Java mapping

Kalm's built-in types map directly to Java's primitive and reference model. Java signatures remain unchanged at the boundary.

| Kalm type | Java representation | Meaning |
|---|---|---|
| `Unit` | `void` return only | No result |
| `Bool` | `boolean` / `Boolean` boxed | `true` or `false` |
| `Byte` | `byte` / `Byte` boxed | Signed 8-bit integer |
| `Short` | `short` / `Short` boxed | Signed 16-bit integer |
| `Int` | `int` / `Integer` boxed | Signed 32-bit integer |

| | | |
|---|---|---|
| `Long` | `long` / `Long` boxed | Signed 64-bit integer |
| `Float` | `float` / `Float` boxed | IEEE 754 binary32 |
| `Double` | `double` / `Double` boxed | IEEE 754 binary64 |
| `Char` | `char` / `Character` boxed | One UTF-16 code unit |
| `Text` | `java.lang.String` | Immutable Unicode text |
| `Any` | `java.lang.Object` | Explicit Java-compatible top type |
| `Array<T>` | `T[]` | Fixed-size reference array; primitive components boxed |
| `Task<T>` | `kalm.runtime.Task<T>`, a Java subtype of `CompletionStage<T>` | Asynchronous result handle |
| `BoolArray` ... `CharArray` | Matching Java primitive array | Fixed-size primitive arrays, including `ShortArray` for Java `short[]` |
| `List<T>` | `java.util.List<T>` | Ordered mutable list |
| `Map<K,V>` | `java.util.Map<K,V>` | Mutable map with non-null keys |
| `T?` | Boxed/reference type | Nullable value; `null` is its null value |

`Unit` is allowed only as a function or method result. Nullable primitive types use Java wrapper classes. List literals `[a, b]` create mutable `java.util.ArrayList` values; map literals create insertion-ordered `java.util.LinkedHashMap` values. Generic arguments are invariant in Kalm; Java generic declarations retain their bounds and wildcard information during interop.

Arithmetic requires matching types, except `Byte`, `Short`, and `Char` promote to `Int` as in Java. There are no implicit narrowing conversions. Explicit conversions are `toByte()`, `toShort()`, `toInt()`, `toLong()`, `toFloat()`, `toDouble()`, and `toChar()`. Integral narrowing is checked and throws `ArithmeticException`; floating-to-integral conversion truncates toward zero and rejects NaN, infinity, and out-of-range values. Integer overflow wraps at the result width. Integer division truncates toward zero.

`true` and `false` are Boolean literals. `null` requires a nullable target or `Any`. A nullable receiver uses safe navigation (`value?.name`). Safe indexing uses `value?[index]`; it evaluates the receiver once and, if it is null, returns null without evaluating the index. The receiver must be nullable. It is supported for arrays, `Text`, `List`, and `Map`; the result is nullable, and a missing map key also yields null. The null-coalescing operator `left ?? right` evaluates `left` once and evaluates `right` only when `left` is null. Its left operand must be nullable, and the result type is the common type of the non-null left type and the right type (and is nullable when the right type is nullable). ??= assigns a fallback only when a mutable nullable local or field is null; the target and fallback are each evaluated at most once. As an expression, ??= yields the existing non-null value or the newly assigned fallback. These are syntax-level operations and add no runtime library dependency. Inside `if (value != null) { ... }`, a stable `fix` local or field is refined to non-null. Mutable `var` values are not smart-cast. Values of type Any require `as Type` before specialized use; failed casts raise `ClassCastException`. `is Type` checks and refines a stable value.

Local `fix`/`var` types may be inferred when initialized. Parameters, fields, and public function results require explicit types. `Task<T>` is a public Kalm runtime interface that extends `java.util.concurrent.CompletionStage<T>`. Generated Java methods keep `Task<T>` as their return

type, and Java callers can use or assign it as a `CompletionStage<T>`. `await` accepts either `Task<T>` or any Java `CompletionStage<T>`. An empty collection or `null` requires a target type. Kalm lambdas require a target type from context and may target Kalm or Java single-abstract-method contracts (SAMs).

# 4. Bindings, expressions, and functions

`fix` declares an immutable binding; `var` declares a mutable one. In type or member modifier position, `fix` means that a class cannot be subclassed or a member cannot be overridden. Fields use the same binding words. Definite assignment is checked at compile time.

KALM

```
fix startValue: Int = 3;
var score = startValue + 4;
score += 1;

fun square(value: Int) -> Int = value * value;
fun greet(name: Text) -> Text {
    return @"Hey, $name!";
}
```

`fun name(...)` declares a named function or method; `fun item => expression` introduces a context-typed lambda expression. Named functions are allowed in declaration positions and use `fun name(...)`; lambda expressions appear in expression positions and use `fun parameters => ...` or `fun parameters { ... }`. Function parameters are immutable. A block-bodied function returning a value must `return` a value on every path. A function without a result arrow returns `Unit` if block-bodied. Expression-bodied functions use `= expression`. A bare `return` is valid only in a `Unit` function. Kalm functions cannot be overloaded, and calls do not support default or named arguments.

Operators evaluate left to right. `and` and `or` short-circuit. Equality uses primitive value equality, text value equality, and Java `equals` for other references. Relational comparisons support numbers, `Char`, and `Text` in UTF-16 order. `+` adds numbers or joins two `Text` values. Operators `&&`, `||`, and `!` are accepted as aliases for `and`, `or`, and `not`. Assignment targets must be mutable Kalm bindings/fields or assignable Java fields. List and array indexes use Java bounds behavior. Safe navigation evaluates its receiver once and skips arguments when the receiver is `null`.

| Precedence high to low | Operators | Associativity |
|---|---|---|
| Postfix | call, member `.`, safe member `?.`, safe index `?[...]`, index `[]` | left |
| Unary | `not`, `!`, unary `+`, unary `-` | right |
| Multiplicative | `*`, `/`, `%` | left |
| Additive | `+`, `-` | left |
| Relational/type | `<`, `<=`, `>`, `>=`, `is`, `as` | non-associative |
| Equality | `==`, `!=` | non-associative |
| Logical | `and`, `or` | left, short-circuit |

| | | |
|---|---|---|
| Coalescing | ?? | right |
| Assignment | =, +=, -=, *=, /=, %=, ??= | right |

A method call to Java uses Java overload resolution. If overload resolution is ambiguous, compilation fails and the diagnostic identifies the candidate signatures. An explicit type annotation or cast can resolve the ambiguity. Kalm does not support operator overloading.

```kalm
var names: List<Text>? = findNames();
fix firstName = names?[0] ?? "Guest";

var title: Text? = findTitle();
title ??= "Untitled";
```

# 5. Flow, lambdas, and errors

`if`, `elif`, and `else` branch on `Bool` conditions; `while` repeats as long as its condition is true. `for item in items` accepts `Iterable<T>` or an array and creates an immutable loop variable. `stop` exits the innermost loop; `next` advances to its next iteration.

```kalm
if (score > 0) {
    sayLine("Ready.");
} elif (score == 0) {
    sayLine("Waiting for input.");
} else {
    sayLine("Let's reset.");
}

while (score < 10) {
    score += 1;
}

for (name in names) {
    sayLine(name);
}
```

Java SAM lambdas use a lightweight Kalm form. Their target and parameter types come from the selected method signature, and captures follow Java's effectively-final rule.

```kalm
fix loud: java.util.function.Function<Text, Text> = fun name => name.toUpperCase();
fix loudNames = names.stream().map(fun name => name.toUpperCase()).toList();
```

A block lambda uses braces: `fun item { ... }`; a value-returning lambda uses `return` to provide its result. The compiler rejects an ambiguous SAM overload unless a target type or cast resolves it. Named functions use `fun name(...)`; lambdas use `fun parameters => expression` or a braced lambda body.

Kalm uses JVM exceptions. `try`, `handle`, `finally`, and `raise` map directly to Java. A `handle` parameter must have a Java exception subtype. A `try` block must have at least one `handle` or a `finally` clause.

Checked Java exceptions must be handled at Kalm call sites; ordinary Kalm functions do not declare `throws` clauses. When an `override fun`, accessor, `forward`, or `external fun` corresponds to a Java method with checked exceptions, generated Java preserves the compatible Java `throws` clause. `raise` may rethrow unchecked exceptions freely; a checked exception may be raised only within a `try` whose `handle` clause catches it. A checked exception cannot escape an ordinary Kalm function. Other unhandled exceptions behave as they do in Java.

KALM

```
try {
    sayLine(Files.readString(Path.of("greeting.txt")));
} handle (error: java.io.IOException) {
    sayLine(error.getMessage());
} finally {
    sayLine("done");
}
```

## Matching and checks

`choice` declares a finite set of named variants and compiles to a Java enum; variants carry no payload or type parameters in 1.1. `choose` evaluates a value once and selects a `when` branch. It must be exhaustive for a `choice` and for a `closed` hierarchy; _ is the explicit catch-all branch.

KALM

```
choice Status {
    Ready;
    Failed;
}

fix message = choose status {
    when Status.Ready => "Ready to continue";
    when Status.Failed => "Please retry";
};
```

`expect condition` raises `AssertionError` when false. `require condition` validates a function precondition and raises `IllegalArgumentException`; `ensure condition` validates a postcondition and raises `IllegalStateException`. Optional messages follow a comma. These checks are always active and are not removed by build settings.

## Resource and scope management

`with name = expression { ... }` requires an `AutoCloseable` result and closes it when the block exits, including exceptional exits. Multiple resources close in reverse declaration order. `defer { ... }` registers a cleanup block for the end of the current function or block; deferred blocks run in reverse registration order on normal return or exception. Control transfer (`return`, `stop`, or `next`) is not allowed inside a deferred block.

KALM

```
with reader = Files.newBufferedReader(path) {
    return reader.readLine();
}

defer {
    transaction.rollbackIfOpen();
```

```kalm
}
```

### Tasks and synchronization

`async fun` runs a function body on the Kalm virtual-thread executor and returns `Task<T>`, whose public Java interface extends `java.util.concurrent.CompletionStage<T>`. Its declared result type `T` is the value produced by the body. `await task` waits for completion and returns its value; it is permitted only inside an `async fun` or a `task { ... }` block. Awaiting rethrows runtime exceptions and errors, and wraps checked failures in `TaskFailureException` with the original cause. Interruption restores the interrupt flag and raises `TaskInterruptedException`.

`task { ... }` starts an asynchronous block and returns `Task<T>`. Its body follows ordinary function block rules: it must return a value on every path, using `return` for a value result. A `guard lock { ... }` block uses the Java monitor for `lock`; `await` is prohibited inside a guard so a task never suspends while holding a monitor. Each `async fun` or `task` executes on a Java virtual thread.

```kalm
async fun loadProfile(id: Text) -> Profile {
    return await profileClient.fetch(id);
}

fix refresh = task {
    return await loadProfile("42");
};

guard cache {
    cache.put("profile", refresh);
}
```

The public `kalm.runtime.Task<T>` interface extends `CompletionStage<T>` and adds `cancel()` and `isCancelled()`. Cancelling a task requests cancellation and may interrupt its virtual thread. Java methods that return `Task<T>` remain directly usable as `CompletionStage<T>`; `await` also accepts ordinary Java completion stages.

## 6. Classes, contracts, and visibility

`class` declares a JVM class. `contract` declares a JVM interface. `init` declares a constructor. If none is present, the compiler follows the default-constructor rule below. `self` refers to the current instance. Calling `ClassName(arguments)` constructs Kalm and Java classes. A Kalm class may declare at most one constructor; named factory functions can provide additional construction forms. Fields initialize in declaration order. A `fix` field is assigned exactly once in its declaration or constructor. Every non-null reference field must be initialized or definitely assigned by every constructor; nullable references default to `null`, and uninitialized primitive `var` fields use their Java zero value. A no-argument constructor is generated only when all required fields can be initialized without parameters; otherwise a missing constructor is a compile error.

```kalm
pub class Person {
    fix name: Text;
    var mood: Text;
```

```kalm
    init(name: Text) {
        self.name = name;
        self.mood = "easy";
    }

    fun hello() -> Text = @"Hi, I'm $name!";
}

contract Greeter {
    fun greet(name: Text) -> Text;
}

class FriendlyGreeter impl Greeter {
    override fun greet(name: Text) -> Text = @"Hey, $name. Glad you're here.";
}
```

Kalm classes compile to Java classes and contracts compile to Java interfaces. A class may `inherits` one `base`, `closed`, or `extendable` Kalm class (only an explicitly permitted child may inherit a `closed` class), or any accessible non-final Java class. It may `impl` one or more contracts or Java interfaces.

A `base class` is abstract and inheritable. Only a `base class` may declare `needs fun` members without bodies; concrete subclasses must implement them. A `needs fun` declaration is an abstract requirement and is implicitly overridable until implemented. A concrete implementation must use `override fun`; it satisfies the requirement even though ordinary methods are closed to overriding by default. An implementing member is itself closed unless marked `extendable`. Contract signatures are also abstract requirements: implementing a Kalm contract or Java interface requires `override fun`, and the resulting method is closed unless marked `extendable`.

A `closed class` must declare every allowed direct child as a nested declaration; the compiler emits a Java sealed hierarchy with an explicit permits list. Nested Kalm classes are static nested Java types; nested classes do not capture an enclosing instance. In a closed hierarchy, a direct child with no modifier or `fix` is Java `final`; an `extendable` child is Java `non-sealed`; and a `closed` child is Java `sealed` and declares its own permits list. A `base` child is abstract and is also emitted as `non-sealed` unless combined with `closed`.

Kalm classes and methods are closed to inheritance or overriding by default. `fix` maps to Java `final`; `extendable` leaves a class or method open; `base` maps to `abstract`; and `closed` maps to Java `sealed`. `base closed` is allowed. `fix` cannot be combined with `base`, `closed`, or `extendable`; `closed` cannot be combined with `extendable`. An override may be marked `fix` to close that method again. `override fun` must match the inherited signature after generic substitution.

`data` declares an immutable Java record with components, accessors, and Java record value equality. `choice` declares a finite set of named variants and compiles to a Java enum; variants carry no payload or type parameters in 1.1. `singleton` declares one globally accessible instance, generated as a final Java class with a private constructor and `INSTANCE` field. `alias Name = Type` creates a compile-time type alias and no new JVM type. `given` adds bounds to type parameters, for example `fun sort<T>(items: List<T>) -> List<T> given T: Comparable<T>`. The clause is required when a type parameter has a bound.

`generate Display` overrides a data record's generated `toString()` with a stable `TypeName[field=value, ...]` form, using component declaration order and Kalm text formatting for each value. Java record equality and hashing remain those provided by the record model; they are not separately generated. `augment Type { ... }` declares Kalm-only extension functions. Inside an augment function,

`self` is the receiver value; calls are rewritten to generated static helpers whose first parameter is that receiver, and do not add methods to the Java class. `bridge augment` also emits a public Java helper facade for those functions. `forward Contract to field` generates contract methods that delegate to that field; the field must implement the contract. Generated forwarding methods preserve the Java contract's checked-exception declarations.

KALM

```
use java.io.Closeable;
use java.io.InputStream;

pub data User(name: Text, id: Long) generate Display;

alias UserId = Long;

augment Text {
    fun initials() -> Text = self.substring(0, 1);
}

class FileStore impl Closeable {
    fix stream: InputStream;

    init(stream: InputStream) {
        self.stream = stream;
    }

    forward Closeable to stream;
}
```

`read` and `write` declare property accessors. Kalm member access rewrites to the accessor; Java receives conventional `getX`/`setX` methods. `lazy fix` declares a read-only field evaluated once on first access and safely publishes the cached value. The initializer runs at most once, including when it throws; subsequent reads rethrow the same failure. `late var` is a non-null reference field assigned after construction; reads from both Kalm and Java use a runtime guard that throws `IllegalStateException` until assignment. Setters reject null, and subsequent assignments are allowed. The compiler does not verify that a `late` field is assigned before use. `constant` declares a public static final Java compile-time constant restricted to primitive or `Text` literal values; the compiler emits a Java constant variable where Java permits it. Constants are implicitly `pub` and top-level in 1.1. `legacy` marks a declaration deprecated in Kalm diagnostics and emits Java `@Deprecated`.

KALM

```
use java.time.ZoneId;

class Profile {
    var first: Text;
    var last: Text;

    init(firstName: Text, lastName: Text) {
        first = firstName;
        last = lastName;
    }

    read fullName: Text = @"$first $last";
    write displayName(value: Text) {
        fix pieces = value.split(" ");
        first = pieces[0];
```

```kalm
        last = pieces[1];
    }

    lazy fix zone: ZoneId = ZoneId.systemDefault();
}

constant DEFAULT_LIMIT: Int = 100;
```

Visibility words map to Java access as closely as Java permits:

| Kalm word | Meaning |
|---|---|
| `pub` | Public to Java and other modules |
| `internal` | Current Kalm module; enforced by Kalm compilation. Java callers may see generated public members because JVM bytecode has no Kalm-module visibility. |
| `private` | Source-file private at top level and class-private for members. Java has no private top-level declarations, so the compiler enforces file privacy and emits package-private Java declarations; same-package Java source can therefore access those generated declarations. |
| `protected` | Java protected visibility for class members |
| omitted | Top-level declarations are `internal`; fields are `private`; methods and constructors are `pub` |

Kalm classes do not declare static members. Java static members remain directly accessible.

# 7. Packages and Java interoperability

`space` gives a Java-compatible package name. `use` imports a Java or Kalm type; wildcard uses are supported. Name resolution checks local declarations, same package, explicit uses, wildcard uses, then `java.lang`. Ambiguity is an error and can be resolved by a qualified name.

Kalm can use public Java classes, constructors, fields, methods, enums, arrays, and generic types found on the compile classpath or readable module path. Kalm primitive names map to matching Java primitives. Calls follow Java visibility, overload resolution, primitive widening, boxing/unboxing, and varargs rules. Narrowing needs an explicit checked conversion. Java annotations describing nullability are honored; unannotated Java reference types are platform types and produce a warning on assignment to a non-null Kalm type.

Java and Kalm declarations in the same module may call one another. Java code can call public top-level Kalm functions through the Java facade for their source file (for example, `GreetingApiKalm.greeting("Tim")`); Kalm code can call accessible Java declarations in the same module. Java and Kalm declarations are resolved together, while compiler-generated analysis artifacts are not visible to source code.

```kalm
space studio.greetings;
use java.nio.file.Files;
use java.nio.file.Path;
```

```kalm
pub fun readGreeting() -> Text {
    try {
        return Files.readString(Path.of("greeting.txt"));
    } handle (error: java.io.IOException) {
        return "Could not read greeting";
    }
}
```

`external fun` declares a Kalm-named wrapper around a public Java static method. Its right-hand qualified name is resolved against the project's Java model, and its signature must be invocation-compatible with the target. Java libraries themselves remain directly usable without wrappers.

```kalm
external fun parseCount(value: Text) -> Int = java.lang.Integer.parseInt;
```

`bridge augment` emits a Java helper facade for Kalm extension functions. An ordinary `augment` is usable only from Kalm; a bridged one also exposes static methods such as `TextExtensions.initials(value)` for Java callers.

`annotation` declares a Java annotation type. `target` lists where it may appear (`type`, `function`, `field`, `property`, or `parameter`). Java has no `ElementType.PROPERTY`: a Kalm `property` target maps to `ElementType.METHOD` and is emitted on the generated getter; use `field` to annotate the backing field. Other targets map to the matching Java `ElementType`. Annotation members use Java's supported annotation value types, and generated annotations default to `RetentionPolicy.CLASS`; Java meta-annotations such as `@Retention`, `@Target`, and `@Repeatable` may configure them. `@Name` or `@Name(arguments)` applies an annotation to a declaration. Parameters use `@Name parameter: Type`; type uses use `parameter: @Name Type`, including nested generic and array types. Type parameters and record components accept annotations. The target shorthand also supports `constructor`, `localVariable`, `annotationType`, `typeUse`, `typeParameter`, and `recordComponent`; conflicting shorthand and explicit `@Target` are rejected. Arguments must be Java annotation constant values, enum constants, class literals, arrays, or nested annotations; positional syntax is allowed only for a single member named `value`. `legacy` emits `@Deprecated` and a Kalm compiler warning at use sites.

A `test "description" { ... }` block declares one test case. A failed `expect` check fails that test. File-level `before { ... }` and `after { ... }` hooks run before and after each test in the file; `after` runs even when a test fails. `expect` checks are always active inside tests.

```kalm
before {
    TestDatabase.openShared();
}

test "a user can be loaded" {
    fix user = TestSupport.repository().find("42");
    expect user.name == "Ari";
}

after {
    TestDatabase.closeShared();
}
```

# 8. Standard library

The standard prelude is automatically available in every Kalm source file. Prelude functions follow the same overload resolution and generic rules as Java methods, and declarations in the current Kalm scope take precedence over prelude names.

Kalm runtime types include `Task<T>`, `TaskFailureException`, and `TaskInterruptedException`. Kalm uses Java APIs for collections, I/O, time, mathematics, and concurrency.

| Kalm API | Behavior |
|---|---|
| `say(value)`, `sayLine(value)` | Write formatted value, optionally followed by the platform newline |
| `len(value)` | Length/size of `Text`, arrays, `List`, or `Map` |
| `arrayOf(values...)` | Construct a reference array when component type is reifiable |
| `boolArrayOf` ... `charArrayOf` | Construct the matching Java primitive arrays |
| `range(end)`, `range(start, end)` | Lazy `Iterable<Int>`; exclusive end |
| `exit(code)` | Delegate to `System.exit` |
| `abs`, `min`, `max` | Numeric overloads for `Int`, `Long`, `Float`, and `Double` |
| `sqrt`, `pow`, `round` | Delegate to `java.lang.Math` |

Formatting uses Java-compatible `null`, `true`, and `false` spellings and formats other values through `String.valueOf`. Interpolation uses the same formatting. `upper()` and `lower()` use locale-independent `Locale.ROOT`. Lists and maps are Java mutable collections. Java APIs such as `Objects`, `Collections`, `Files`, and `Math` remain directly importable.

# 9. Core grammar

This grammar is normative. Semicolons or logical newlines terminate simple statements; braces or colon suites delimit blocks, as defined in Section 2. A semantic modifier check rejects duplicate modifiers and the invalid combinations described in Section 6. The parser rejects missing statement terminators, unmatched braces, and malformed statements rather than guessing. The precedence productions below resolve the shorthand `Expr` rule used in statement productions. ??= is a right-associative assignment operator; ?? is a right-associative coalescing operator between logical and assignment precedence.

EBNF

```
File              = [ SpaceDecl ] { UseDecl } { Declaration } EOF ;
SpaceDecl         = "space" QualifiedName Terminator ;
UseDecl           = "use" QualifiedName [ "as" Identifier | "." "*" ] Terminator ;
Declaration       = { AnnotationUse } [ Visibility ] [ "legacy" ]
                    ( TypeDecl | FunctionDecl | AliasDecl | ConstantDecl
                    | AugmentDecl | ExternalDecl | AnnotationDecl | TestDecl | HookDecl ) ;
AnnotationUse     = "@" QualifiedName [ "(" [ AnnotationArguments ] ")" ] ;
AnnotationArguments = AnnotationArgument { "," AnnotationArgument } ;
AnnotationArgument  = [ Identifier "=" ] AnnotationValue ;
AnnotationValue     = ConstantExpr | Type "." "class" | "{" [ AnnotationValues ] "}" |
```

EBNF

```
NestedAnnotation ;
AnnotationValues    = AnnotationValue { "," AnnotationValue } ;
NestedAnnotation    = "@" QualifiedName [ "(" [ AnnotationArguments ] ")" ] ;
Visibility        = "pub" | "internal" | "private" | "protected" ;
TypeDecl          = ClassDecl | ContractDecl | DataDecl | ChoiceDecl | SingletonDecl ;
TypeModifier      = "base" | "closed" | "fix" | "extendable" ;
ClassDecl         = { TypeModifier } "class" Identifier [ TypeParams ] [ InheritsClause ] [
ImplClause ] MemberBlock ;
InheritsClause    = "inherits" Type ;
ImplClause        = "impl" TypeList ;
ContractDecl      = "contract" Identifier [ TypeParams ] [ "inherits" TypeList ]
ContractBlock ;
DataDecl          = "data" Identifier [ TypeParams ] "(" [ Params ] ")" [ GenerateClause ]
Terminator ;
ChoiceDecl        = "choice" Identifier ChoiceBlock ;
ChoiceBlock       = "{" { ChoiceVariant } "}" | ":" DeclarationSuite(ChoiceVariant) ;
ChoiceVariant     = Identifier Terminator ;
SingletonDecl     = "singleton" Identifier MemberBlock ;
MemberBlock       = "{" { Member } "}" | ":" DeclarationSuite(Member) ;
Member            = { AnnotationUse } [ Visibility ] [ "legacy" ]
                    ( Field | Constructor | FunctionDecl | NeedsFunction
                    | Accessor | ForwardDecl | TypeDecl ) ;
ContractBlock     = "{" { ContractMember } "}" | ":" DeclarationSuite(ContractMember) ;
ContractMember    = { AnnotationUse } ( FunctionSignature | AccessorSignature ) ;
FunctionHead      = [ "async" ] "fun" Identifier [ TypeParams ] "(" [ Params ] ")"
                    [ "->" Type ] [ GivenClause ] ;
FunctionSignature = FunctionHead Terminator ;
NeedsFunction     = "needs" FunctionSignature ;
Constructor       = "init" "(" [ Params ] ")" Block ;
FunctionDecl      = [ "override" ] [ "async" ] [ "extendable" | "fix" ] "fun" Identifier [
TypeParams ]
                    "(" [ Params ] ")" [ "->" Type ] [ GivenClause ] FunctionBody ;
FunctionBody      = "=" Expr Terminator | Block ;
AliasDecl         = "alias" Identifier [ TypeParams ] "=" Type Terminator ;
ConstantDecl      = "constant" Identifier ":" Type "=" ConstantExpr Terminator ;
Field             = [ "lazy" ] [ "late" ] ( "fix" | "var" ) Identifier ":" Type [ "=" Expr ]
Terminator ;
Accessor          = "read" Identifier ":" Type ( "=" Expr Terminator | Block )
                    | "write" Identifier "(" Param ")" Block ;
AccessorSignature = "read" Identifier ":" Type Terminator
                    | "write" Identifier "(" Param ")" Terminator ;
ForwardDecl       = "forward" Type "to" Identifier Terminator ;
GenerateClause    = "generate" Identifier { "," Identifier } ;
GivenClause       = "given" TypeConstraint { "," TypeConstraint } ;
TypeConstraint    = TypeParam ":" Type ;
TypeList          = Type { "," Type } ;
Params            = Param { "," Param } ;
Param             = { AnnotationUse } Identifier ":" Type ;
TypeParams        = "<" TypeParam { "," TypeParam } ">" ;
TypeParam         = { AnnotationUse } Identifier ;
Type              = { AnnotationUse } NamedType [ "?" ] ;
NamedType         = QualifiedName [ "<" Type { "," Type } ">" ]
                    | "Unit" | "Bool" | "Byte" | "Short" | "Int" | "Long"
                    | "Float" | "Double" | "Char" | "Text" | "Any" | "Task" "<" Type ">"
                    | "Array" "<" Type ">" | "BoolArray" | "ByteArray" | "ShortArray"
                    | "IntArray" | "LongArray" | "FloatArray" | "DoubleArray" | "CharArray"
                    | "List" "<" Type ">" | "Map" "<" Type "," Type ">" ;
```

EBNF

```
AugmentDecl       = [ "bridge" ] "augment" Type AugmentBlock ;
AugmentBlock      = "{" { AugmentFunction } "}" | ":" DeclarationSuite(AugmentFunction) ;
AugmentFunction   = [ Visibility ] [ "legacy" ] FunctionDecl ;
ExternalDecl      = "external" "fun" Identifier [ TypeParams ] "(" [ Params ] ")"
                    [ "->" Type ] [ GivenClause ] "=" QualifiedName Terminator ;
AnnotationDecl    = "annotation" Identifier AnnotationBlock ;
AnnotationBlock   = "{" { AnnotationMember } "}" | ":" DeclarationSuite(AnnotationMember) ;
AnnotationMember  = TargetDecl | AnnotationElement ;
TargetDecl        = "target" TargetKind { "," TargetKind } Terminator ;
TargetKind        = "type" | "function" | "field" | "property" | "parameter"
                    | "constructor" | "localVariable" | "annotationType" | "typeUse"
                    | "typeParameter" | "recordComponent" ;
AnnotationElement = Identifier ":" Type [ "=" AnnotationValue ] Terminator ;
TestDecl          = "test" TextLiteral Block ;
HookDecl          = ( "before" | "after" ) Block ;
Block             = "{" { Statement } "}" | ":" Suite ;
Suite             = SimpleSuite | NEWLINE INDENT Statement { Statement } DEDENT ;
SimpleSuite       = SimpleStatement { SimpleStatement } ;
DeclarationSuite(Element) = Element { Element }
                    | NEWLINE INDENT Element { Element } DEDENT ;
SimpleStatement   = Binding | ReturnStmt | RaiseStmt | CheckStmt | StopStmt | NextStmt
                    | Expr Terminator ;
BlockEnd          = "}" | DEDENT ;
Terminator        = ";" | NEWLINE | BlockEnd | EOF ;
Statement         = Binding | IfStmt | WhileStmt | ForStmt | ChooseStmt | ReturnStmt |
RaiseStmt
                    | TryStmt | WithStmt | DeferStmt | GuardStmt | CheckStmt | StopStmt |
NextStmt
                    | Block | Expr Terminator ;
Binding           = { AnnotationUse } ( "fix" | "var" ) Identifier [ ":" Type ] "=" Expr
Terminator ;
IfStmt            = "if" Condition Block { "elif" Condition Block } [ "else" Block ] ;
Condition         = Expr | "(" Expr ")" ;
WhileStmt         = "while" Condition Block ;
ForStmt           = "for" ( "(" Identifier "in" Expr ")" | Identifier "in" Expr ) Block ;
ChooseStmt        = ChooseExpr Terminator ;
ChooseExpr        = "choose" Expr ChooseBlock ;
ChooseBlock       = "{" { WhenArm } "}" | ":" DeclarationSuite(WhenArm) ;
WhenArm           = "when" Pattern "=>" ( Expr Terminator | Block ) ;
Pattern           = QualifiedName | "_" | Literal ;
ReturnStmt        = "return" [ Expr ] Terminator ;
RaiseStmt         = "raise" Expr Terminator ;
TryStmt           = "try" Block ( HandleClause { HandleClause } [ "finally" Block ]
                    | "finally" Block ) ;
HandleClause      = "handle" "(" Identifier ":" Type ")" Block ;
WithStmt          = "with" Resource { "," Resource } Block ;
Resource          = Identifier "=" Expr ;
DeferStmt         = "defer" Block ;
GuardStmt         = "guard" Expr Block ;
CheckStmt         = ( "expect" | "require" | "ensure" ) Expr [ "," Expr ] Terminator ;
StopStmt          = "stop" Terminator ;
NextStmt          = "next" Terminator ;
TaskExpr          = "task" Block ;
Lambda            = "fun" LambdaParams "=>" Expr | "fun" LambdaParams Block ;
LambdaParams      = LambdaParam { "," LambdaParam } | "(" [ LambdaParam { "," LambdaParam } ]
")" ;
LambdaParam       = { AnnotationUse } Identifier [ ":" Type ] ;
```

EBNF

```
Expr              = Assignment ;
Assignment        = Coalesce [ AssignmentOp Assignment ] ;
Coalesce          = LogicalOr [ "??" Coalesce ] ;
AssignmentOp      = "=" | "+=" | "-=" | "*=" | "/=" | "%=" | "??=" ;
LogicalOr         = LogicalAnd { ( "or" | "||" ) LogicalAnd } ;
LogicalAnd        = Equality { ( "and" | "&&" ) Equality } ;
Equality          = Relational [ ( "==" | "!=" ) Relational ] ;
Relational        = Additive [ ( "<" | "<=" | ">" | ">=" ) Additive | ( "is" | "as" ) Type ]
;
Additive          = Multiplicative { ( "+" | "-" ) Multiplicative } ;
Multiplicative    = Unary { ( "*" | "/" | "%" ) Unary } ;
Unary             = ( "not" | "!" | "+" | "-" | "await" ) Unary | Postfix ;
Postfix           = Primary { Call | MemberAccess | SafeMember | SafeIndex | Index } ;
Call              = [ TypeArgs ] "(" [ Arguments ] ")" ;
MemberAccess      = "." Identifier ;
SafeMember        = "?." Identifier ;
SafeIndex         = "?[" Expr "]" ;
Index             = "[" Expr "]" ;
Arguments         = Expr { "," Expr } ;
Primary           = Literal | Identifier | "self" | "(" Expr ")" | List | Map | Lambda |
TaskExpr | ChooseExpr ;
List              = "[" [ Arguments ] "]" ;
Map               = "{" [ MapEntry { "," MapEntry } ] "}" ;
MapEntry          = Expr ":" Expr ;
Literal           = IntegerLiteral | FloatLiteral | CharLiteral | TextLiteral |
InterpolatedText
                    | "true" | "false" | "null" ;
ConstantExpr      = Literal | QualifiedName | Unary | Additive ;
TypeArgs          = "<" Type { "," Type } ">" ;
QualifiedName     = Identifier { "." Identifier } ;
```

Calls on a name resolving to a type construct it; other calls invoke a function or method. Generic type arguments before constructor parentheses support Java calls such as `ArrayList<Text>()`. `choose` patterns match enum constants, closed-hierarchy types, literals, or _; destructuring is not supported in 1.1. Braced blocks are valid only in declarations, control flow, resource/task/guard blocks, tests, and lambdas. `fix` has context-dependent but related meanings: immutable binding, non-extendable class, or non-overridable member. A value introduced by `fix` cannot be reassigned. `fun` introduces both named functions and lambdas: declarations have a name and parenthesized parameters, while lambdas use => or a brace/colon body. Colon lambdas require parenthesized parameters. Expression parsing follows the precedence table in Section 4.

# 10. Diagnostics

A diagnostic reports a stable code, severity, UTF-16 source range, concise message, and optional related locations. Codes use `KALM-1xxx` for syntax errors, `KALM-2xxx` for type and name errors, `KALM-3xxx` for Java interoperability errors, and `KALM-9xxx` for compiler errors. Syntax, type, and interoperability errors prevent compilation.

# References


- Java Language Specification, Java SE 25
- Java Virtual Machine Specification, Java SE 25

- Virtual threads, JDK 25
- CompletableFuture, JDK 25
- Record classes, JDK 25